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Apparatus, methods and computer programs for reducing echo

Abstract

Apparatus, methods and computer programs are provided for reducing echo in microphone signals. The apparatus can be configured to obtain an echo path delay estimate and obtain information indicative of a number of future frames used by an acoustic echo cancellation filter. The apparatus can also be configured to obtain information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter. The apparatus can also be configured to determine at least a first delay to be added to a loudspeaker signal and determine at least a second delay to be added to a microphone signal. The determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Great Britain Application No. 2118594.7, filed Dec. 21, 2021, the entire contents of which are incorporated herein by reference.

TECHNOLOGICAL FIELD

(2) Examples of the disclosure relate to apparatus, methods and computer programs for reducing echo. Some relate to apparatus, methods and computer programs for reducing echo in microphone signals.

BACKGROUND

(3) Audio capture systems can enable immersive voice applications such as teleconferencing or mediated reality environments. Such systems use simultaneous audio capture and playback to enable real time spatial audio to be transmitted over communications networks. Acoustic echo cancellation is used in such systems to remove acoustic echoes of the loudspeakers from the microphone signals. This avoids the loudspeaker echo degrading the quality of the spatial audio captured by the microphones.

BRIEF SUMMARY

(4) According to various, but not necessarily all, examples of the disclosure there is provided an apparatus comprising means for: obtaining an echo path delay estimate; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter; determining at least a first delay to be added to a loudspeaker signal; and determining at least a second delay to be added to a microphone signal; wherein the determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay is obtained that is dependent upon the time shift.

(5) The first delay and the second delay may be used to provide a reduced echo microphone signal.

(6) Reducing echo may comprise one or more of: increasing echo return loss enhancement; increasing short time objective intelligibility score; and increasing perceptual evaluation of speech quality score

(7) The first delay and the second delay may reduce processing latency of the microphone signal.

(8) The reduction in processing latency may comprise reducing the processing latency between a microphone and an output of an acoustic echo cancellation processing block.

(9) The time-frequency filterbank upon which the acoustic echo cancellation filter is based may comprise at least one of: short term Fourier transform filterbank; weighted overlap-add filterbank;

in combination with one or more sub-band adaptive acoustic echo cancellation filters.

(10) The effective echo path delay may comprise an echo path delay estimate adjusted by the first delay and the second delay.

(11) The time shift between consecutive frames processed by the acoustic echo cancellation filter may comprise a hop size and the effective echo path delay comprises an integer number of hop sizes.

(12) The means may be for enabling the first delay to be provided to a first delay unit where the first delay unit is configured to add a delay to at least one loudspeaker signal.

(13) The means may be for enabling the first delay to be provided to a plurality of first delay units where the plurality of first delay units are configured to add a delay to a plurality of loudspeaker signals.

(14) The means may be for enabling the second delay to be provided to a second delay unit where the second delay unit is configured to add a delay to at least one microphone signal.

(15) The means may be for enabling the second delay to be provided to a plurality of second delay units where the plurality of second delay units are configured to add a delay to a plurality of microphone signals.

(16) The means may be for enabling a plurality of third delays to be added to a plurality of reduced echo signals to enable encoding for spatial audio.

(17) The means may be for enabling the same second delay to be provided to a plurality of second delay units where the plurality of second delay units are configured to add a delay to a plurality of microphone signals.

(18) The means may be for enabling the first delay to be added in the time domain and the second delay to be added in the time domain.

(19) The means may be for enabling the second delay to be added, at least partially, in the time domain and, at least partially, in the frequency domain.

(20) According to various, but not necessarily all, examples of the disclosure there is provided an apparatus comprising at least one processor; and at least one memory including computer program code, the at least one memory and the computer program code configured to, with the at least one processor, cause the apparatus at least to perform: obtaining an echo path delay estimate; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter; determining at least a first delay to be added to a loudspeaker signal; and determining at least a second delay to be added to a microphone signal; wherein the determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay is obtained that is dependent upon the time shift.

(21) According to various, but not necessarily all, examples of the disclosure there is provided an electronic device comprising an apparatus as claimed in any preceding claim wherein the electronic device is at least one of: a telephone, a camera, a computing device, a teleconferencing device.

(22) According to various, but not necessarily all, examples of the disclosure there is provided a method comprising: obtaining an echo path delay estimate; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter; determining at least a first delay to be added to a loudspeaker signal; and determining at least a second delay to be added to a microphone signal; wherein the determination of the first delay and the second delay is based on

the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay is obtained that is dependent upon the time shift.

(23) According to various, but not necessarily all, examples of the disclosure there is provided a computer program comprising computer program instructions that, when executed by processing circuitry, cause: obtaining an echo path delay estimate; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter; determining at least a first delay to be added to a loudspeaker signal; and determining at least a second delay to be added to a microphone signal; wherein the determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay is obtained that is dependent upon the time shift.

Description

BRIEF DESCRIPTION

- (1) Some examples will now be described with reference to the accompanying drawings in which:
- (2) FIG. 1 shows an example system;
- (3) FIG. 2 shows an echo path impulse response;
- (4) FIG. 3 shows an example method;
- (5) FIG. 4 shows an example system;
- (6) FIG. 5 shows a plot relating ERLE to channel delay;
- (7) FIG. 6 shows a plot of microphone signals and delayed microphone signals;
- (8) FIGS. 7A to 7D show plots of results obtained using examples of the disclosure;
- (9) FIG. 8 shows an example system;
- (10) FIG. 9 shows an example system;
- (11) FIG. 10 shows an example system;
- (12) FIG. 11 shows an example system; and
- (13) FIG. 12 shows an example apparatus.

DETAILED DESCRIPTION

(14) FIG. 1 shows an example system; **101** in which examples of the disclosure could be implemented. The system **101** comprises at least one loudspeaker **103** and at least one microphone **105**. The system **101** could be provided within an electronic device such as a mobile phone, a camera, a teleconferencing device, a computing device or any other suitable type of device.

(15) The system **101** is configured so that a loudspeaker signal **107** is provided to the loudspeaker **103**. This controls the loudspeaker **103** to generate audio. The microphones **105** can comprise any means that can be configured to detect sounds and convert the sounds to an electrical output microphone signal **111**.

(16) In this system **101** an echo path **109** exists between the loudspeaker **103** and the microphones **105**. This means that audio from the loudspeaker **103** can be detected by the microphones **105**. This can create an unwanted echo within the microphones signals **111** provided by the microphones.

(17) The example system **101** shown in FIG. 1 comprises a single loudspeaker **103** and a single microphone **103**. In other examples the system **101** could comprise a plurality of loudspeakers **103** and/or a plurality of microphones **105**.

(18) Examples of the disclosure can be used in such systems to reduce the unwanted echo.

(19) FIG. 2 shows an example echo path impulse response. The echo path impulse response could be for a system **101** such as the system shown in FIG. 1. This is the response captured at the microphone when the loudspeaker is fed with an impulse signal (with a non-zero value at time zero and a zero value elsewhere).

(20) The echo path impulse response comprises a dead time **201** and an active region **203**. As shown in FIG. 2 there is no echo response in the dead time **201**. The dead time **201** can be indicative of the time delay of the true physical echo path. During the dead time **201** the channel output from the microphone **105** will be silent.

(21) Different components and aspects of the system **101** can contribute to the dead time **201**. In the example system of FIG. 1 the dead time **201** can comprise a system delay component and a propagation time component. The propagation time component is dependent on the echo path **109** between the loudspeaker **103** and the microphone **105**. The system component is dependent upon delays such as data buffering by the audio device, latency within components such as analogue to digital converters (ADCs) and digital to analogue converters (DACs) or other relevant delays within the system **101**.

(22) The channel output from the microphone **105** is not silent during the active region **203**. The active region **203** comprises an impulse response **205** a plurality of secondary peaks **207** and reverberation tail **209**. The impulse response **205** corresponds to a line-of-sight peak for the echo path. This can also be referred to as the direct sound. The plurality of secondary peaks **207** correspond to the early reflections. The early reflections are typically related to the geometry of the acoustic environment. The reverberation tail **209** corresponds to late reflections that have a more diffuse nature.

(23) Different shapes of echo path impulse responses could be generated in different systems **101** and examples.

(24) In examples of the disclosure an acoustic echo cancellation filter can be applied to the loudspeaker signal **107** and the filtered loudspeaker signal can then be combined with the microphone signals to help to reduce the echo within the microphone signals **111**. The acoustic echo cancellation filter can be designed to fit the echo path impulse response as well as possible.

(25) The acoustic echo cancellation filter can be designed to comprise a delay corresponding to the dead time **201**. There is no need to filter this part of a signal because the echo path impulse response is zero-valued in the dead time **201** region. The acoustic echo cancellation filter can be shifted using a delay so that it only covers the active region **203**. The use of the delay can reduce the overall length of the acoustic echo cancellation filter that is needed. It can also increase the amount of the reverberation tail **209** that is covered by the acoustic echo cancellation filter and so can improve the reduction of the echo. Reducing the overall length of the acoustic echo cancellation filter can provide for a faster start-up time and can reduce computational complexity.

(26) Examples of the disclosure provide for a method of reducing the echo within audio systems such as the system **101** of FIG. 1 by controlling the delays that are added to the microphone signals **111** and the loudspeaker signals **109**. In some examples these delays could be optimised, or substantially optimised, so as to reduce echo within microphone signals **111**.

(27) FIG. 3 shows an example method that can be implemented in examples of the disclosure. The method could be implemented in a system **101** such as the system shown in FIG. 1. The method could be used to determine the delay that is to be added to the loudspeaker signal **107** and the microphone signal **111** so as to reduce the effects of echo in the system **101**. FIG. 3 shows an order to the blocks. Implementations of the disclosure could apply the blocks of the method in any suitable order.

(28) At block **301** the method comprises obtaining an estimate for an echo path delay. The echo path delay comprises components due to the propagation of the audio along the echo path **109** between the loudspeaker **103** and the microphone **105** and also components due to processing

delays within the system **101**.

(29) At block **303** the method comprises obtaining filter parameters. The filter parameters can be parameters relating to an acoustic echo cancellation filter that can be used to filter a loudspeaker signal **107** before it is combined with a microphone signal **111**.

(30) The filter parameters that are obtained can comprise information indicative of a number of future frames used by the acoustic echo cancellation filter to generate an acoustic echo cancelled signal for a current time frame.

(31) The acoustic echo cancellation filter can comprise any suitable type of filter. The acoustic echo cancellation filter can comprise a time-frequency filterbank. The time-frequency filterbank upon which the acoustic echo cancellation filter is based can comprise any suitable type of filterbank. For example, it could comprise a short-term

(32) Fourier transform filterbank, a weighted overlap-add filterbank or any other suitable type of filterbank. The filterbank can be provided in combination with one or more sub-band adaptive acoustic echo cancellation filters.

(33) At block **305** the method comprises obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter. The information can be indicative of a time shift between any consecutive time frames. The time shift between consecutive frames processed by the acoustic echo cancellation filter comprises a hop size or any other suitable parameter. The hop size can be the number of samples between consecutive frames of the filterbank. The consecutive frames can be partially overlapping in time.

(34) At block **307** the method comprises determining at least a first delay to be added to the loudspeaker signal **107** and at block **309** the method comprises determining at least a second delay to be added to the microphone signal **11**.

(35) The determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter. Examples of how the first delay and the second delay can be calculated are described herein.

(36) The first delay and the second delay can be determined so that the effective echo path delay is dependent upon the time shift between consecutive frames processed by the acoustic echo cancellation filter. The effective echo path delay comprises the echo path delay estimate adjusted by the first delay and the second delay. In some examples first delay and the second delay can be determined so that the effective echo path delay can comprise an integer number of hop sizes of the filterbanks of the acoustic echo cancellation filter.

(37) In examples of the disclosure the first delay and the second delay adjust the effective echo path delay so as provide a reduced echo microphone signal. The reduced echo microphone signal comprises less echo than the microphone signal. The echo could be cancelled, or substantially cancelled, in the reduced echo microphone signal. Reducing echo can comprise controlling any suitable parameters. In some examples reducing echo can comprise, increasing echo return loss enhancement, increasing short time objective intelligibility score, increasing perceptual evaluation of speech quality score and/or controlling any other parameter. The echo return loss enhancement can comprise the ratio of the microphone power and the residual echo. In some examples controlling the parameter can comprise maximising or substantially maximising the appropriate parameter.

(38) In some examples the sizes of the first delay and the second delay can be selected so as to reduce processing latency of the microphone signal **111**. The reduction in the processing latency comprises reducing the processing latency between a microphone **103** and an output of an acoustic echo cancellation processing block provided within a system **101**.

(39) Once the method of FIG. 3 has been performed the first delay can be added to a loudspeaker signal **107** and the second delay can be added to a microphone signal **111**. To enable the first delay to be added to a loudspeaker signal **107** information indicative of the first delay can be provided to

one or more first delay units. This can enable the first delay unit to add the appropriate delay to the loudspeaker signal **107**. Similarly, to enable the second delay to be added to a microphone signal **111** information indicative of the second delay can be provided to one or more second delay units. This can enable the second delay unit to add the appropriate delay to the microphone signal **111**.

(40) In examples where the system **101** comprises a plurality of loudspeakers **103** a plurality of first delay units can be comprised within the system **101**. In some examples the different first delay units can be configured to add different first delays as appropriate to the different loudspeaker signals **107**. Similarly in some examples the system **101** could comprise a plurality of microphones **103**. In such examples the system **101** can comprise a plurality of second delay units configured to second delays as appropriate to the different microphone signals **111**. In some examples different second delays could be added to different microphone signals **111**. In some examples the same second delays could be added to the different microphone signals **111**.

(41) The delays can be added to the microphone signals **111** and the loudspeaker signals **107** in any suitable domain. For example, both the first delay and the second delay can be added in the time domain. In some examples the second delay can be added, at least partially in the time domain and, at least partially, in the frequency domain.

(42) In the example of FIG. **3** a first delay and a second delay are determined. In other examples additional delays could be determined. For instance, in some examples a third delay could be added after the acoustic echo cancellation block in order to synchronise signals for spatial audio encoding or for any other suitable purpose.

(43) FIG. **4** schematically shows an example system **101** that can be used to implement methods of the disclosure such as the method shown in FIG. **3**. The system **101** comprises a loudspeaker **103** and a microphone **105**. A loudspeaker signal **107** is provided to the loudspeaker **103** and a microphone signal **111** is provided by the microphone **105**. An echo path **109** exists between the loudspeaker **103** and the microphones **105**.

(44) In the example of FIG. **4** the system **101** also comprises an acoustic echo cancellation block **401**. The acoustic echo cancellation block **401** comprises means for enabling delays to be added to microphone signal **111** and the loudspeaker signal **107** to reduce the echo in the microphone signal **111**. The output of the acoustic echo cancellation block **401** can comprise an echo-reduced microphone signal **111**.

(45) In this example the acoustic echo cancellation block **401** comprises a delay estimation block **403**, delay controller **407**, a first delay unit, **413**, a first analysis filter bank **415**, sub-band adaptive filters **417**, a second delay unit **419** and a second analysis filterbank **421**. The acoustic echo cancellation block **401** could comprise different components and/or have different configurations in other examples of the disclosure.

(46) The delay controller block **407** is configured to implement an algorithm that defines how parameters such as the echo path delay estimate, the number of future frames of an acoustic echo cancellation filter and the time shift between consecutive frames processed by the acoustic echo cancellation filter are used to determine the delays that are to be added to the microphone signal **111** and the loudspeaker signal **107**. The delay controller block **407** receives a delay estimation from the delay estimation block **403** as an input.

(47) The delay controller block **407** also receives an input **405** indicative of the number of future frames of an acoustic echo cancellation filter. The delay controller block can also receive an input indicative of the time-shift between consecutive frames of the acoustic echo cancellation filter. The acoustic echo cancellation filter can be used to filter the loudspeaker signal **107**. The future frames can be referred to as lookahead frames.

(48) The delay controller **407** can comprise any means that can be configured to use the delay inputs from the delay estimation algorithm block **403** and the input **405** indicative of the number of future frames of an acoustic echo cancellation filter, and an input indicative of the time-shift between consecutive frames of the acoustic echo cancellation filter, to determine a first delay for

the loudspeaker signal **107** and a second delay for the microphone signal **111**. The delay controller **407** can use the method of FIG. **3**, or any other suitable method, to determine the first delay and the second delay. The delay controller **407** can comprise an apparatus **1201** as shown in FIG. **12** and/or any other suitable means.

(49) The delay controller **407** provides a first output **409** comprising information indicative of the first delay and a second output **411** comprising information indicative of the second delay.

(50) The first output **409** is provided to the first delay unit **413**. The first delay unit **413** can comprise any means that can be configured to add a delay into a signal. The first delay unit **413** is configured to enable a delay to be added to a loudspeaker signal **107**. The loudspeaker signal **107** comprises a reference of the echo that is to be reduced or cancelled.

(51) The first delay unit **413** is configured to enable the delay to be added to the loudspeaker signal **107** before the loudspeaker signal **107** is filtered. The first output **409** controls the size of the delay that is added to the loudspeaker signal **107**.

(52) The first delay unit **413** provides a delayed loudspeaker signal as an output. The delayed loudspeaker signal is provided to an acoustic echo cancellation filter. The acoustic echo cancellation filter can comprise any means that can be configured to remove the echo impulse response from the loudspeaker signal **107**. In this example the acoustic echo cancellation filter comprises an analysis filterbank **415** in combination with one or more sub-band adaptive acoustic echo cancellation filters **417**.

(53) The analysis filterbank **415** can comprise a short-term Fourier transform filterbank, a weighted overlap-add filterbank or any other suitable type of filterbank. The filterbanks are configured to split the acoustic echo cancellation filter into a plurality of short-parallel frequency domain sub-bands. The sub-band adaptive acoustic echo cancellation filters **417** can then be used to filter the sub-bands at a reduced sampling rate.

(54) The second output **411** of the delay controller **407** is provided to the second delay unit **419**. The second delay unit **419** can comprise any means that can be configured to add a delay into a microphone signal **111**. The microphone signal **111** comprises an altered version of the loudspeaker signals that is to be reduced or cancelled and a desired signal that is to be retained.

(55) The second delay unit **419** is configured to enable a delay to be added to a microphone signal **111**. In the example of FIG. **4** the second delay unit **419** is configured to enable the delay to be added to the microphone signal **111** before the microphone signal **111** is filtered. In other examples the second delay unit **419** can be configured to enable the delay to be added to the microphone signal **111** after the microphone signal **111** is filtered by the filterbank **421** but before the acoustic echo cancellation block **401**. The second output **411** controls the size of the delay that is added to the microphone signal **111**.

(56) The second delay unit **419** provides a delayed microphone signal **111** as an output. The delayed microphone signal is provided to a second analysis filterbank **421**. The second analysis filterbank **421** can comprise the same type of filterbank as the first filterbank **415**. For example, the second analysis filterbank **421** could comprise a short-term Fourier transform filterbank, a weighted overlap-add filterbank or any other suitable type of filterbank.

(57) The acoustic echo cancellation block **401** is configured to enable the filtered microphone signal to be combined with the filtered loudspeaker signal. This can enable the echo to be reduced or cancelled within the microphone signal. The acoustic echo cancellation block **401** therefore provides an output **423** comprising a reduced echo microphone signal.

(58) The reduced echo microphone signal can be provided to a synthesis filter bank **425** to enable rendering of audio based on the microphone signals. The synthesis filter bank **425** can enable spatial audio rendering or any other suitable type of rendering.

(59) The combination of the echo path delay and the first delay added to the loudspeaker signal and the second delay added to the microphone signal result in an effective echo path delay. The effective echo path delay $T_{sub.D}$ can be defined as:

$$T_{\text{sub.D}} = T_{\text{sub.D}} - T_{\text{sub.1}} + T_{\text{sub.2}}$$

(60) Where $T_{\text{sub.D}}$ is the dead time **201** of the echo path impulse response $T_{\text{sub.1}}$ is the value of the first delay that is added to the loudspeaker signal **107** $T_{\text{sub.2}}$ is the value of second delay that is added to the microphone signal **111**.

(61) This equation shows the case for a single loudspeaker **103** and a single microphone **105**. In other examples a plurality of loudspeakers **103** and/or microphones **105** can be provided within the system **101**. In such cases an effective echo path delay $T_{\text{sub.D.sup.km}}$ can be determined for the different pairings of loudspeakers k and microphones m .

(62) The first delay therefore effectively reduces the echo path delay and the second delay effectively increases the echo path delay.

(63) Systems **101** which have a large dead time **201** $T_{\text{sub.D}}$ induce a delay in the microphone signal **111**. This effectively generates lookahead or future frames for the acoustic echo cancellation filter. The effect of this is shown in FIG. **5** where having a larger dead time **201** $T_{\text{sub.D}}$ enables a higher echo return loss enhancement to be obtained. Therefore, if a system **101** has a large dead time **201** $T_{\text{sub.D}}$ this can be taken into account by the delay controller **407**. For example, the delay controller **407** can reduce the processing latency of the filters.

(64) In examples of the disclosure the values of the first delay and the second delay can be controlled so as to minimise:

$$(65) \min_{T_1, T_2} |T_D - T_1 + T_2 - k * L| \text{ .Math. } ^2$$

(66) Where k is the number of future frames used by the acoustic echo cancellation filter and L is a hop size of the filterbank. The hop size of the filterbank provides information indicative of the time shift between consecutive frames processed by the acoustic echo cancellation filter. This minimisation can optimise, or substantially optimise, the echo reduction in the system. This minimisation can also minimise, or substantially minimise, the latency caused by the delays and the processing of the signals within the system **101**.

(67) FIG. **5** shows a plot relating echo return loss enhancement (ERLE) to dead time **201** which can also be referred to as the echo path time delay $T_{\text{sub.D}}$. The ERLE is a ratio of the microphone power and the residual echo within the microphone signal **111**.

(68) The data used for the plot in FIG. **5** was obtained from a simulation with a measured room impulse response (100 ms long). The AEC filter used was WOLA-NLMS (Weighted Overlap-Add Normalized Least mean Square) with 66% overlap (3× oversampling). The AEC filter was used with the first delay $T_{\text{sub.1}}$ equal to 0 and the second delay $T_{\text{sub.2}}$ equal to $\{0, \text{hop-size}, 2 \times \text{hop-size}\}$, which corresponds to a number of $k = \{0, 1, 2\}$ lookahead filter frames respectively, The hop size of the filterbank was **320** samples and a fixed NLMS step size was used. The sampling rate used was 48 kHz. White noise was used as the input signal and the ERLE was measured after convergence.

(69) This plot shows that peaks of maximum ERLE are obtained if the echo path time delays combined with the second delay $T_{\text{sub.2}}$ are close to $k * \text{hop-size}$ where k is an integer. That is, a maximum ERLE can be obtained if the effective echo path delay $T_{\text{sub.D}}$ is an integer number of hop sizes or is close to an integer number of hop sizes.

(70) The plot also shows that steep drops or minima of ERLE occur if the echo path time delays are close to $(1/2 + k) * \text{hop-sizes}$. That is, a minimum ERLE occurs if the effective echo path delay $T_{\text{sub.D}}$ is an odd integer number of half-hop sizes or is close to an odd integer number of half-hop sizes.

(71) It can be seen in FIG. **5** that the ERLE increases when having a larger second delay $T_{\text{sub.2}}$. That is, having a larger number of future frames or lookahead frames for the acoustic echo cancellation filter can provide significant gains for ERLE however this comes at the cost of increased processing latency.

(72) FIG. **6** shows plots of microphone signals and delayed microphone signals. The first plot **601**

shows a microphone signal **111** that has not been delayed. This can be the case where there is no further processing of the microphone signals **111**

(73) The second plot **603** shows latency savings that can be obtained using examples of the disclosure. The second plot **603** shows an example output of an acoustic echo cancellation block that uses examples of this disclosure. The third plot **605** shows latency savings that can be obtained for systems that do not use examples of the disclosure. The third plot **605** shows an example output of an acoustic echo cancellation block that does not use examples of this disclosure. The acoustic echo cancellation block that does not use examples of the disclosure could use a SOTA delay scheme.

(74) In this case the dead time **201** is equal to $L/2$ where L is a hop-size of the filterbank. A twice-oversampled WOLA-AEC (Weighted Overlap Add Acoustic Echo cancellation) filter (with 50% overlap) and one lookahead, or future, filter frame was used.

(75) In this case the minimum possible latency of the filterbank is one hop-size which is equal to L samples. This is the inherent latency of the filterbank which is independent of any lookahead, or future, filter frames.

(76) In the example which does not use examples of the disclosure the delay scheme involves removing the dead time **201** by delaying the loudspeaker signal provided to the acoustic echo cancellation filter. This adds a delay equal to the dead time **201** of $L/2$. A delay is then added to the microphone signal **111**. This delay is equal to one hop size L . The delay added to the microphone signal **111** enables lookahead, or future, filter frames to be used. This delay can be added in the frequency domain. This creates a total processing latency of $2L$ samples.

(77) In the scenario that does use examples of the disclosure the first delay that is added to the loudspeaker signal **107** can be set to zero so that $T_{sub.1}=0$. The second delay that is added to the microphone signal **111** can be set to $L/2$ so that $T_{sub.2}=L/2$. The effective echo path delay $T_{sub.D}$ is therefore $T_{sub.D}=L$. This enables one lookahead or future frame to be used while only introducing half of the additional latency. In this example the total processing latency is $1.5L$ samples.

(78) The example of the disclosure therefore reduces the processing latency by $0.5L$ samples. For a sampling rate of 48 kHz and $L=480$ samples, this saves 5 ms.

(79) Reducing the latency provides an improved Quality of Experience (QoE) for the user. The latency that is relevant for the QoE is the latency experienced by the microphone signal **111** from the moment it is generated by the microphone **105** to the moment that it is rendered to an audio signal on the other end of the call. Any latency reduction in the acoustic echo cancellation processing will therefore provide an improved QoE.

(80) FIGS. 7A to 7D show plots of results obtained using examples of the disclosure. These show improvements in latency that can be obtained using examples of the disclosure.

(81) FIG. 7A plots the dead time **201** in units of hop size of the filterbank on the x axis and the additional latency introduced by the acoustic echo cancellation processing block on the y axis. The first plot **701** shows the latency for systems **101** which do not implement examples of the disclosure while the second plot **703** shows the latency for systems that for implement examples of the disclosure. The point **705** in the second plot shows the situation for the scenario given in FIG. 6.

(82) FIG. 7B shows the values for the first delay $T_{sub.1}$ that is to be added to the loudspeaker signal **107** in the first plot **707** and the second delay $T_{sub.2}$ that is to be added to the microphone signal **111** in the second plot **709**. The x axis plots the dead time **201** in units of hop size of the filterbank and they axis plots the additional latency introduced by the acoustic echo cancellation processing block.

(83) In the example of FIGS. 7A and 7B one lookahead filter frame, or future filter frame, is used by the acoustic echo filter.

(84) FIGS. 7C and 7D show similar plots to those in FIGS. 7A and 7B however in the examples of

FIGS. 7C and 7D two lookahead filter frames, or future filter frames, are used by the acoustic echo filter.

(85) In FIG. 7C the first plot **711** shows the latency for systems **101** which do not implement examples of the disclosure while the second plot **713** shows the latency for systems that for implement examples of the disclosure. Similarly in FIG. 7D the values for the first delay $T_{sub.1}$ that is to be added to the loudspeaker signal **107** in the first plot **715** and the second delay $T_{sub.2}$ that is to be added to the microphone signal **111** in the second plot **717**.

(86) These plots show that using the examples of disclosure and selecting appropriate delays can provide for reduced latency compared to systems that do not use examples of the disclosure.

(87) FIG. **8** shows an example system **101** that comprises a plurality of loudspeakers **103** and a single microphone **103**. The different loudspeakers **103** can have different dead times **201** between the respective loudspeakers **103** and the microphones **105**.

(88) This system **101** comprises an acoustic echo cancellation block **401** similar to the acoustic echo cancellation block **401** shown in FIG. **4**. Corresponding reference numerals are used for corresponding components.

(89) To enable the different dead times **201** to be taken into account the delay controller **407** can be configured to provide a plurality of first outputs **409A**, **409B**. The plurality of first outputs **409A**, **409B** comprise information indicative of a first delay that is to be added to a loudspeaker signal **107**. Different outputs **409A**, **409B** can indicate different sized delays. This can enable delays of different sizes to be added to different loudspeaker signals **107**.

(90) The first outputs **409A**, **409B** are provided to first delay units **413A**, **413B**. In FIG. **8** only two first delay unit **413A**, **413B** are shown. The number of first units **413** could be the same as the number of loudspeakers **103**. A different first delay unit **413** could be provided for each loudspeaker **103**. Each of the first delay units **413** are configured to enable a different delay to be added to a loudspeaker signal **107**.

(91) The first delay unit **413** provides a delayed loudspeaker signal as an output. The delayed loudspeaker signal is provided to an acoustic echo cancellation filter. In this example a different acoustic echo cancellation filter is provided for each microphone signal **111**. In this example each acoustic echo cancellation filter comprises an analysis filterbank **415A**, **415B** in combination with one or more sub-band adaptive acoustic echo cancellation filters **417A**, **417B**.

(92) The second output **411** of the delay controller **407** is provided to the second delay unit **419** to enable a delay to be added to the microphone signal **111**. The delayed microphone signal is provided to a second analysis filterbank **421**. The acoustic echo cancellation block **401** is configured to enable the filtered microphone signal to be combined with the filtered loudspeaker signals.

(93) The system in FIG. **8** is configured so that the reduced echo microphone signal can be provided to a synthesis filter bank **425** to enable rendering of audio based on the microphone signals. The synthesis filter bank **425** can enable spatial audio rendering or any other suitable type of rendering. The synthesized signal can then be used to provide spatial audio **427** such as OZO spatial audio or any other suitable type of audio.

(94) The delay controller **407** can use any suitable process and/or algorithm to determine the delays that are to be added by the respective delay units **413**, **419**. In some examples the following algorithm could be used:

(95) TABLE-US-00001 INPUT: L Hop-size $[T_{sub.D}]_{sub.l}$ Echo path time delay between loudspeaker l and (reference) microphone B Lookahead filter frames OUTPUT: $[T_{sub.1}]_{sub.l}$ first delay that is to be added to a loudspeaker signal 409 $T_{sub.2}$ second delay that is to be added to the microphone signal 411 ALGORITHM:

$$l_{min} \leftarrow \min_l [T_D]_{l1} \quad \begin{array}{l} \% \text{find shortest time delay between all loudspeakers} \\ \text{and reference microphone} \end{array} \quad \text{If}$$

[T.sub.D].sub.l.sub.min.sub.1 > B * L [T.sub.1].sub.l = [T.sub.D].sub.l - B * L; T.sub.2 = 0; Else [T.sub.1].sub.l.sub.min = 0; [T.sub.1].sub.l = [T.sub.D].sub.l - [T.sub.D].sub.l.sub.min for l ≠ l.sub.min T.sub.2 = B * L - [T.sub.D].sub.l.sub.min; end

(96) FIG. 9 schematically shows another example system **101** that can be used to implement examples of the disclosure. In this example the system **101** can comprise a single loudspeaker **103** and a plurality of microphones **105**. This system **101** could be used for spatial audio capture. The different microphones **105** can have different dead times **201** between the respective loudspeakers **103** and the microphones **105**.

(97) This system **101** also comprises an acoustic echo cancellation block **401** similar to the acoustic echo cancellation block **401** shown in FIGS. 4 and 8. Corresponding reference numerals are used for corresponding components.

(98) To enable the different dead times **201** to be taken into account the delay controller **407** can be configured to provide a plurality of second outputs **411A**, **411B**. The plurality of second outputs **411A**, **411B** comprise information indicative of a second delay that is to be added to a microphone signal **111**. Different outputs **411A**, **411B** can indicate different sized delays. This can enable delays of different sizes to be added to different microphone signals **111**.

(99) In the example of FIG. 9 a first output **409** is provided to a first delay unit **419**. The first delay unit **413** provides a delayed loudspeaker signal as an output. The delayed loudspeaker signal is provided to an acoustic echo cancellation filter. In this example a different acoustic echo cancellation filter is provided for each microphone signal **111A**, **111B**.

(100) The plurality of second outputs **411A**, **411B** are provided to different second delay units **419A**, **419B**. In FIG. 9 only two second delay units **419A**, **419B** are shown. The number of second delay units **419A**, **419B** could be the same as the number of microphones **105**. A different second delay unit **419** could be provided for each microphone **105**. Each of the second delay units **419A**, **419B** are configured to enable a different delay to be added to a microphone signal **111**.

(101) The second delay units **419A**, **419B** are configured to enable delays to be added to the different microphone signals **111A**, **111B**. The delayed microphone signals are provided to second analysis filterbanks **421A**, **421B**. In this example a different filterbank **421A**, **421B** is provided for each of the microphone signals **111**. This can enable the delays to be optimised, or substantially optimised, for the different microphones **105**.

(102) The acoustic echo cancellation block **401** is configured to enable the filtered loudspeaker signals to be combined with each filtered microphone signal. This provides a plurality of separate reduced echo microphone signals **423A**, **423B**. In this example a different reduced echo microphone signal **423A**, **423B** is provided for each of the microphones **105A**, **105B** in the system **101**.

(103) The system **101** of FIG. 9 also comprises a synthesis filter bank **425A**, **425B** for each of the microphones **105A**, **105B** in the system **101**. Each of the reduced echo microphone signals **423A**, **423B** are provided to a different synthesis filter bank **425A**, **425B**.

(104) The system **101** of FIG. 9 is configured to add a third delay to be added to the reduced echo microphone signals **423A**, **423B** after they have been processed by the synthesis filterbanks **425A**, **425B**. In this system **101** the delay controller **407** is configured to provide a third output **901**. The third output **901** comprises information indicative of a third delay.

(105) The third output **901** is provided to the third delay units **903A**, **903B**. The third delay units **903A**, **903B** can comprise any means that can be configured to add a delay into a reduced echo microphone signal **423A**, **423B** after they have been processed by the synthesis filterbanks **425A**, **425B**.

(106) In the example of FIG. 9 a plurality of third delay units **903A**, **903B** are provided. The plurality of third delay units **903A**, **903B** can be configured to enable different third delays to be added to different reduced echo microphone signals **423A**, **423B**.

(107) The third delay can be configured to enable synchronization of the reduced echo microphone

signals **423A**, **423B**. The synchronization realigns the timings of the reduced echo microphone signals **423A**, **423B** so as to enable encoding for spatial audio.

(108) The third delay units **903A**, **903B** provide a synchronised reduced echo microphone signals **905A**, **905B** as an output. These can then be used to provide spatial audio **427** such as OZO spatial audio or any other suitable type of audio.

(109) The third delay that is added accounts for the different delays that are added to the different microphone signals **111A**, **111B** by the different second delay units **419A**, **419B**. This additional third delay will result in an increase in latency for the system **101** however this is likely to be small if the microphones **105** are fairly close to each other. However, this slight increase in latency will provide improved spatial audio quality due to the improved echo reduction and realignment of the signals.

(110) In some examples the third delay $T_{\text{sub.3}}$ can be determined by the smallest second delay $T_{\text{sub.2}}$. For examples the third delay $T_{\text{sub.3.sup.n}}$ that is to be added for microphone signal **105 n** can be determined by:

$$(111) T_3^n = \max_m \{ T_2^m \} - T_2^n$$

(112) FIG. **10** schematically shows another example system **101** that can be used to implement examples of the disclosure. In this example the system **101** also comprises a single loudspeaker **103** and a plurality of microphones **105**. This system **101** could also be used for spatial audio capture. The different microphones **105** can have different dead times **201** between the respective loudspeakers **103** and the microphones **105**. The system in FIG. **10** is different to the system of FIG. **9** in that the same second delay is added to each of the microphone signals **111A**, **111B**. the different dead times **201** between the respective loudspeakers **103** and the microphones **105** can be accounted for by adding different first delays to the loudspeaker signal **107**.

(113) This system **101** shown in FIG. **10** comprises an acoustic echo cancellation block **401** similar to the acoustic echo cancellation block **401** shown in FIGS. **4**, **8** and **9**. Corresponding reference numerals are used for corresponding components.

(114) To enable the different dead times **201** to be taken into account the delay controller **407** can be configured to provide a plurality of first outputs **409A**, **409B**. The plurality of first outputs **409A**, **409B** comprise information indicative of a first delay that is to be added to a loudspeaker signal **107**. Different outputs **409A**, **409B** can indicate different sized delays. This can enable delays of different sizes to be added to the loudspeaker signals **107**. These different delays compensate for the different dead times **201** between the loudspeakers **103** and the microphones **105**.

(115) The first delay units **413A**, **413B** provide delayed loudspeaker signals as an output. The delayed loudspeaker signals are provided to an acoustic echo cancellation filter. In this example the acoustic echo cancellation filter comprises an analysis filterbank **415A**, **415B** for each of the delayed loudspeaker signals and a single sub-band adaptive acoustic echo cancellation filter **417**.

(116) The second output **411** of the delay controller **407** is provided to different second delay units **419A**, **419B**. In FIG. **9** only two second delay units **419A**, **419B** are shown. The number of second delay units **419A**, **419B** could be the same as the number of microphones **105**. A different second delay unit **419** could be provided for each microphone **105**. In this example each of the second delay units **419A**, **419B** are configured to enable the same delay to be added to a microphone signal **111**.

(117) The delayed microphone signals are provided from the second delay units **419A**, **419B** to second analysis filterbanks **421A**, **421B**. In this example a different filterbank **421A**, **421B** is provided for each of the microphone signals **111**.

(118) The acoustic echo cancellation block **401** is configured to enable the filtered loudspeaker signals to be combined with each filtered microphone signal. This provides a plurality of separate reduced echo microphone signals **423A**, **423B**. In this example a different reduced echo microphone signal **423A**, **423B** is provided for each of the microphones **105A**, **105B** in the system

101.

(119) The system **101** of FIG. **10** also comprises a synthesis filter bank **425A**, **425B** for each of the microphones **105A**, **105B** in the system **101**. Each of the reduced echo microphone signals **423A**, **423B** are provided to a different synthesis filter bank **425A**, **425B**.

(120) In this example there is no need to add an additional third delay because the difference in the dead times **201** has been accounted for by adding different first delays. The output signals from the different synthesis filter bank **425A**, **425B** can then be used to provide spatial audio **427** such as OZO spatial audio or any other suitable type of audio.

(121) The example system of FIG. **10** enables the difference in the dead times **201** to be accounted for without adding any additional latency. However, this system **101** would require increased processing and memory usage due to the complexity of the system **101** and the increased number of filterbanks needed.

(122) FIG. **11** schematically shows another example system **101** that can be used that also comprises a single loudspeaker **103** and a plurality of microphones **105**. In this example the different dead times **201** between the respective loudspeakers **103** and the microphones **105** can be accounted for by assuming that the different microphones **105** are very close to each other.

(123) This system **101** shown in FIG. **11** comprises an acoustic echo cancellation block **401** similar to the acoustic echo cancellation block **401** shown in FIGS. **4**, **8**, **9** and **10**. Corresponding reference numerals are used for corresponding components.

(124) The delay controller **407** provides a first output **409** comprising information indicative of the first delay and a second output **411** comprising information indicative of the second delay.

(125) The first output **409** is provided to the first delay unit **413** to enable a delay to be added to a loudspeaker signal **107**. The first delay unit **413** provides a delayed loudspeaker signal as an output. The delayed loudspeaker signal is provided to an acoustic echo cancellation filter. The acoustic echo cancellation filter comprises an analysis filterbank **415** in combination with one or more sub-band adaptive acoustic echo cancellation filters **417**.

(126) The second output **411** of the delay controller **407** is provided to a plurality of second delay units **419A**, **419B**. In FIG. **11** only two second delay units **419A**, **419B** are shown. The number of second delay units **419A**, **419B** could be the same as the number of microphones **105**. A different second delay unit **419** could be provided for each microphone **105**. In this example each of the second delay units **419A**, **419B** are configured to enable the same delay to be added to a microphone signal **111**.

(127) The delayed microphone signals are provided from the second delay units **419A**, **419B** to second analysis filterbanks **421A**, **421B**. In this example a different filterbank **421A**, **421B** is provided for each of the microphone signals **111**.

(128) The acoustic echo cancellation block **401** is configured to enable the filtered loudspeaker signals to be combined with each filtered microphone signal. This provides a plurality of separate reduced echo microphone signals **423A**, **423B**. In this example a different reduced echo microphone signal **423A**, **423B** is provided for each of the microphones **105A**, **105B** in the system **101**.

(129) The system **101** of FIG. **11** also comprises a synthesis filter bank **425A**, **425B** for each of the microphones **105A**, **105B** in the system **101**. Each of the reduced echo microphone signals **423A**, **423B** are provided to a different synthesis filter bank **425A**, **425B**.

(130) In this example there is no need to add an additional third delay because it has been assumed that the microphones **105** are close together so that the difference in the dead times **201** is small with respect to the hop size of the filterbanks. For example, if the system **101** could comprise two microphones that are 20 cm apart. In such cases, for a sound wave travelling parallel to the axis between the microphones, the maximum time difference between incident sound waves on the microphones **105** will be 0.6 ms. This corresponds to 28 samples at 48 kHz. A typical hop size would be 320-960 samples (6.6 to 20 ms). In such examples there would be a reduction in the

performance of the acoustic echo cancellation due to the effective echo path delay $T_{sub.D}$ not being an exact integer number of hop sizes. However, this reduction would be negligible because the effective echo path delay $T_{sub.D}$ is still very close to an integer number of hop sizes.

(131) Therefore, the system **101** in FIG. **11** can operate in a similar manner as the system of (132) FIG. **1** which comprises only a single microphone **105**. In the system of FIG. **11** a single reference microphone **105** can be selected and used to determine the second delay that is to be added to all of the microphone signals **111**.

(133) The system of FIG. **11** is less complex than the examples of FIGS. **9** and **10** and uses fewer memory and processing resources.

(134) Other systems **101** and arrangements could be used in other examples of the disclosure. For example, some systems **101** could comprise a plurality of loudspeakers **103** and a plurality of microphones **105**. These systems **101** could use any suitable combination of the arrangements shown in FIGS. **8** to **11** and/or any other suitable arrangements.

(135) FIG. **12** schematically illustrates an apparatus **1201** that can be used to implement examples of the disclosure. In this example the apparatus **1201** comprises a controller **1203**. The controller **1203** can be a chip or a chip-set. In some examples the controller can be provided within a computer or other device that can be configured to provide signals and receive signals and/or datasets.

(136) In the example of FIG. **12** the implementation of the controller **1203** can be as controller circuitry. In some examples the controller **1203** can be implemented in hardware alone, have certain aspects in software including firmware alone or can be a combination of hardware and software (including firmware).

(137) As illustrated in FIG. **12** the controller **1203** can be implemented using instructions that enable hardware functionality, for example, by using executable instructions of a computer program **1209** in a general-purpose or special-purpose processor **1205** that can be stored on a computer readable storage medium (disk, memory etc.) to be executed by such a processor **1205**.

(138) The processor **1205** is configured to read from and write to the memory **1207**. The processor **1205** can also comprise an output interface via which data and/or commands are output by the processor **1205** and an input interface via which data and/or commands are input to the processor **1205**.

(139) The memory **1207** is configured to store a computer program **1209** comprising computer program instructions (computer program code **1211**) that controls the operation of the controller **1203** when loaded into the processor **1205**. The computer program instructions, of the computer program **1209**, provide the logic and routines that enables the controller **1203** to perform the methods illustrated in FIG. **3** The processor **1205** by reading the memory **1207** is able to load and execute the computer program **1209**.

(140) The apparatus **1201** therefore comprises: at least one processor **1205**; and at least one memory **1207** including computer program code **1211**, the at least one memory **1207** and the computer program code **1211** configured to, with the at least one processor **1205**, cause the apparatus **1201** at least to perform: obtaining an echo path delay estimate; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter; determining at least a first delay to be added to a loudspeaker signal; and determining at least a second delay to be added to a microphone signal; wherein the determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay is obtained that is dependent upon the time shift.

(141) As illustrated in FIG. 12 the computer program **1209** can arrive at the controller **1203** via any suitable delivery mechanism **1213**. The delivery mechanism **1213** can be, for example, a machine readable medium, a computer-readable medium, a non-transitory computer-readable storage medium, a computer program product, a memory device, a record medium such as a Compact Disc Read-Only Memory (CD-ROM) or a Digital Versatile Disc (DVD) or a solid state memory, an article of manufacture that comprises or tangibly embodies the computer program **1209**. The delivery mechanism can be a signal configured to reliably transfer the computer program **1209**. The controller **1203** can propagate or transmit the computer program **1209** as a computer data signal. In some examples the computer program **1209** can be transmitted to the controller **1203** using a wireless protocol such as Bluetooth, Bluetooth Low Energy, Bluetooth Smart, 6LoWPan (IP.sub.v6 over low power personal area networks) ZigBee, ANT+, near field communication (NFC), Radio frequency identification, wireless local area network (wireless LAN) or any other suitable protocol.

(142) The computer program **1209** comprises computer program instructions for causing an apparatus **1201** to perform at least the following: obtaining an echo path delay estimate; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter; determining at least a first delay to be added to a loudspeaker signal; and determining at least a second delay to be added to a microphone signal; wherein the determination of the first delay and the second delay is based on the echo path delay estimate, the number of future frames and the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay is obtained that is dependent upon the time shift.

(143) The computer program instructions can be comprised in a computer program **1209**, a non-transitory computer readable medium, a computer program product, a machine readable medium. In some but not necessarily all examples, the computer program instructions can be distributed over more than one computer program **1209**.

(144) Although the memory **1207** is illustrated as a single component/circuitry it can be implemented as one or more separate components/circuitry some or all of which can be integrated/removable and/or can provide permanent/semi-permanent/dynamic/cached storage.

(145) Although the processor **1205** is illustrated as a single component/circuitry it can be implemented as one or more separate components/circuitry some or all of which can be integrated/removable. The processor **1205** can be a single core or multi-core processor.

(146) References to “computer-readable storage medium”, “computer program product”, “tangibly embodied computer program” etc. or a “controller”, “computer”, “processor” etc. should be understood to encompass not only computers having different architectures such as single/multi-processor architectures and sequential (Von Neumann)/parallel architectures but also specialized circuits such as field-programmable gate arrays (FPGA), application specific circuits (ASIC), signal processing devices and other processing circuitry. References to computer program, instructions, code etc. should be understood to encompass software for a programmable processor or firmware such as, for example, the programmable content of a hardware device whether instructions for a processor, or configuration settings for a fixed-function device, gate array or programmable logic device etc.

(147) As used in this application, the term “circuitry” can refer to one or more or all of the following:

(148) (a) hardware-only circuitry implementations (such as implementations in only analog and/or digital circuitry) and

(149) (b) combinations of hardware circuits and software, such as (as applicable):

(150) (i) a combination of analog and/or digital hardware circuit(s) with software/firmware and

(151) (ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions and

(152) (c) hardware circuit(s) and or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (e.g. firmware) for operation, but the software can not be present when it is not needed for operation.

(153) This definition of circuitry applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term circuitry also covers an implementation of merely a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the particular claim element, a baseband integrated circuit for a mobile device or a similar integrated circuit in a server, a cellular network device, or other computing or network device.

(154) The apparatus **1201** as shown in FIG. **12** can be provided within any suitable device. In some examples the apparatus **1201** can be provided within an electronic device such as a mobile telephone, a teleconferencing device, a camera, a computing device or any other suitable device.

(155) The blocks illustrated in FIG. **3** can represent steps in a method and/or sections of code in the computer program **1209**. The illustration of a particular order to the blocks does not necessarily imply that there is a required or preferred order for the blocks and the order and arrangement of the blocks can be varied. Furthermore, it can be possible for some blocks to be omitted.

(156) The term ‘comprise’ is used in this document with an inclusive not an exclusive meaning. That is any reference to X comprising Y indicates that X may comprise only one Y or may comprise more than one Y. If it is intended to use ‘comprise’ with an exclusive meaning then it will be made clear in the context by referring to “comprising only one . . .” or by using “consisting”.

(157) In this description, reference has been made to various examples. The description of features or functions in relation to an example indicates that those features or functions are present in that example. The use of the term ‘example’ or ‘for example’ or ‘can’ or ‘may’ in the text denotes, whether explicitly stated or not, that such features or functions are present in at least the described example, whether described as an example or not, and that they can be, but are not necessarily, present in some of or all other examples. Thus ‘example’, ‘for example’, ‘can’ or ‘may’ refers to a particular instance in a class of examples. A property of the instance can be a property of only that instance or a property of the class or a property of a sub-class of the class that includes some but not all of the instances in the class. It is therefore implicitly disclosed that a feature described with reference to one example but not with reference to another example, can where possible be used in that other example as part of a working combination but does not necessarily have to be used in that other example.

(158) Although examples have been described in the preceding paragraphs with reference to various examples, it should be appreciated that modifications to the examples given can be made without departing from the scope of the claims.

(159) Features described in the preceding description may be used in combinations other than the combinations explicitly described above.

(160) Although functions have been described with reference to certain features, those functions may be performable by other features whether described or not.

(161) Although features have been described with reference to certain examples, those features may also be present in other examples whether described or not.

(162) The term ‘a’ or ‘the’ is used in this document with an inclusive not an exclusive meaning. That is any reference to X comprising a/the Y indicates that X may comprise only one Y or may comprise more than one Y unless the context clearly indicates the contrary. If it is intended to use ‘a’ or ‘the’ with an exclusive meaning then it will be made clear in the context. In some circumstances the use of ‘at least one’ or ‘one or more’ may be used to emphasis an inclusive meaning but the absence of these terms should not be taken to infer any exclusive meaning.

(163) The presence of a feature (or combination of features) in a claim is a reference to that feature or (combination of features) itself and also to features that achieve substantially the same technical effect (equivalent features). The equivalent features include, for example, features that are variants and achieve substantially the same result in substantially the same way. The equivalent features include, for example, features that perform substantially the same function, in substantially the same way to achieve substantially the same result.

(164) In this description, reference has been made to various examples using adjectives or adjectival phrases to describe characteristics of the examples. Such a description of a characteristic in relation to an example indicates that the characteristic is present in some examples exactly as described and is present in other examples substantially as described.

(165) Whilst endeavoring in the foregoing specification to draw attention to those features believed to be of importance it should be understood that the Applicant may seek protection via the claims in respect of any patentable feature or combination of features hereinbefore referred to and/or shown in the drawings whether or not emphasis has been placed thereon.

Claims

1. An apparatus comprising at least one processor; and at least one memory including computer program code, the at least one memory and the computer program code configured to, with the at least one processor, cause the apparatus at least to: obtain an estimate of an echo path delay comprising an estimate of a time delay between a loudspeaker signal and an altered version of the loudspeaker signal comprised in a microphone signal; obtain information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtain information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter, wherein the time shift between consecutive frames processed by the acoustic echo cancellation filter comprises a hop size; determine at least a first delay to be added to a loudspeaker signal for adjusting the echo path delay to reduce an echo in the microphone signal; and determine at least a second delay to be added to a microphone signal for adjusting the echo path delay to reduce the echo in the microphone signal, wherein the determination of the first delay and the second delay is based on (i) the estimate of the echo path delay, (ii) the number of future frames used by the acoustic echo cancellation filter, and (iii) the time shift between consecutive frames processed by the acoustic echo cancellation filter, and wherein the first delay and the second delay are determined so that an effective echo path delay, resulting from addition of the first delay to the loudspeaker signal and addition of the second delay to the microphone signal, is obtained that is shifted closer to an integer number of hop sizes and further from an odd integer number of half hop sizes.

2. An apparatus as claimed in claim 1, wherein the reducing the echo in the microphone signal comprises one or more of: increasing short time objective intelligibility score; and or increasing perceptual evaluation of speech quality score.

3. An apparatus as claimed in claim 1, wherein the first delay and the second delay reduce processing latency of the microphone signal.

4. An apparatus as claimed in claim 3, wherein the reduction in processing latency comprises reducing the processing latency between a microphone and an output of an acoustic echo cancellation processing block.

5. An apparatus as claimed in claim 1, wherein the time-frequency filterbank upon which the acoustic echo cancellation filter is based comprises at least one of: short term Fourier transform filterbank; and weighted overlap-add filterbank; in combination with one or more sub-band adaptive acoustic echo cancellation filters.

6. An apparatus as claimed in claim 1, wherein the effective echo path delay comprises an integer

number of hop sizes.

7. An apparatus as claimed in claim 1, wherein the apparatus is caused to enable the first delay to be provided to a first delay unit where the first delay unit is configured to add a delay to at least one loudspeaker signal.

8. An apparatus as claimed in claim 7, wherein the apparatus is caused to enable the first delay to be provided to a plurality of first delay units, where the plurality of first delay units are configured to add a delay to a plurality of loudspeaker signals.

9. An apparatus as claimed in claim 1, wherein the apparatus is caused to enable the second delay to be provided to a second delay unit, where the second delay unit is configured to add a delay to at least one microphone signal.

10. An apparatus as claimed in claim 9, wherein the apparatus is caused to enable the second delay to be provided to a plurality of second delay units, where the plurality of second delay units are configured to add a delay to a plurality of microphone signals.

11. An apparatus as claimed in claim 1, wherein the apparatus is caused to enable a plurality of third delays to be added to a plurality of reduced echo signals to enable encoding for spatial audio.

12. An apparatus as claimed in claim 1, wherein the apparatus is caused to enable the second delay to be provided to a plurality of second delay units, where the plurality of second delay units are configured to add a delay to a plurality of microphone signals.

13. An apparatus as claimed in claim 1, wherein the apparatus is caused to enable the first delay to be added in the time domain and the second delay to be added in the time domain.

14. An apparatus as claimed in claim 1, wherein the apparatus is caused to enable the second delay to be added, at least partially, in the time domain and, at least partially, in the frequency domain.

15. An electronic device comprising an apparatus as claimed in claim 1, wherein the electronic device is at least one of: a telephone, a camera, a computing device, a teleconferencing device.

16. The apparatus as claimed in claim 1, wherein the first delay and the second delay are determined to minimize a difference between the effective echo path delay and a result of multiplying the hop size by the number of future frames used by the acoustic echo cancellation filter.

17. The apparatus as claimed in claim 1, wherein the at least one memory and the computer program code are configured to, with the at least one processor, further cause the apparatus to: determine whether the estimate of the echo path delay is longer than a result of multiplying the hop size by the number of future frames used by the acoustic echo cancellation filter; responsive to the estimate of the echo path delay being longer than the result of multiplying the hop size by the number of future frames used by the acoustic echo cancellation filter, determine the second delay is equal to zero; and responsive to the estimate of the echo path delay being shorter than the result of multiplying the hop size by the number of future frames used by the acoustic echo cancellation filter, determine the first delay is equal to zero.

18. A method comprising: obtaining an estimate of an echo path delay comprising an estimate of a time delay between a loudspeaker signal and an altered version of the loudspeaker signal comprised in a microphone signal; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter, wherein the time shift between consecutive frames processed by the acoustic echo cancellation filter comprises a hop size; determining at least a first delay to be added to a loudspeaker signal for adjusting the echo path delay to reduce an echo in the microphone signal; and determining at least a second delay to be added to a microphone signal for adjusting the echo path delay to reduce the echo in the microphone signal, wherein the determination of the first delay and the second delay is based on (i) the estimate of the echo path delay, (ii) the number of future frames used by the acoustic echo cancellation filter, and (iii) the

time shift between consecutive frames processed by the acoustic echo cancellation filter, and wherein the first delay and the second delay are determined so that an effective echo path delay, resulting from addition of the first delay to the loudspeaker signal and addition of the second delay to the microphone signal, is obtained that is shifted closer to an integer number of hop sizes and further from an odd integer number of half hop sizes.

19. A non-transitory computer-readable storage medium comprising computer program instructions that, when executed by processing circuitry, cause: obtaining an estimate of an echo path delay comprising an estimate of a time delay between a loudspeaker signal and an altered version of the loudspeaker signal comprised in a microphone signal; obtaining information indicative of a number of future frames used by an acoustic echo cancellation filter wherein the acoustic echo cancellation filter comprises a time-frequency filterbank configured to generate an acoustic echo cancelled signal for a current time frame; obtaining information indicative of a time shift between consecutive frames processed by the acoustic echo cancellation filter, wherein the time shift between consecutive frames processed by the acoustic echo cancellation filter comprises a hop size; determining at least a first delay to be added to a loudspeaker signal for adjusting the echo path delay to reduce an echo in the microphone signal; and determining at least a second delay to be added to a microphone signal for adjusting the echo path delay to reduce the echo in the microphone signal; wherein the determination of the first delay and the second delay is based on (i) the estimate of the echo path delay, (ii) the number of future frames used by the acoustic echo cancellation filter, and (iii) the time shift between consecutive frames processed by the acoustic echo cancellation filter and wherein the first delay and the second delay are determined so that an effective echo path delay, resulting from addition of the first delay to the loudspeaker signal and addition of the second delay to the microphone signal, is obtained that is shifted closer to an integer number of hop sizes and further from an odd integer number of half hop sizes.
